

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	B Tech IT	Date:	23/7/26

Code of Conduct

I, S. Tharun (192421183) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: [Signature]

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

60
100

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Questions:

1. Given:

Available: 45 MHz Company Policy: 25 KHz

To know how many full duplex voice channels can this spectrum support.

$$= 45000 / 50 = 900 \text{ MHz}$$

2.

Given:

Network 560 7 frequency

Channels per cell = Total channel

$$= 560 / 7 = 80$$

3.

Total voice channels = carriers x slot / carriers = 100×8
 $= 800$

4.

Power ratio: $800 / 4 = 20$

$$\text{Ratio in dB} = 10 \times \log_{10}(20) = 13.01 \text{ dB} = 13 \text{ dB}$$

5.

$$C/I \text{ (dB)} = \text{Signal (dBm)} - \text{Interference (dBm)}$$

$$= 92 \text{ dBm} - 60 \text{ dBm} = 32 \text{ dBm}$$

6.

$$\text{Channels} = 42000 \text{ MHz} / 300 \text{ KHz} = 140 / 7$$

$$= 20$$

7.

Transmitter 64 W
exponent 4

1/d⁴ doubling d
= 16/6 64/16

Received power: 4 W

8.

$4 \times 2^2 = 4 \times$ (plus keeping the same SNR)

9.

Total channel clusters = cell size $\times$ channel / cell = 488 = 60

10.

$$v = 100 \text{ km/h} = 27.78 \text{ m/s} \quad 900 \text{ MHz}$$

$$= v \times f / c = 27.78 \times 900 \times 10^6 / 3 \times 10^8 = 83.4 \text{ Hz}$$

11.

$$\text{Slot duration} = \text{Frame time} / \text{Slots} = 5 \text{ ms} / 8 = 0.625 \text{ ms}$$

12.

Spectrum Planners have 24 MHz

$$\text{Carriers} = \text{Total spec} / \text{spacing time}$$

$$= 24000 / 200$$

$$= 120$$

13.

Handset Transmits at 8 W, signal experience 33 dB

$$= 34.03 \times 33 = 6.03 \text{ dB} \quad \log(8000) \text{ mV}$$

$$= 6 \text{ dB}$$

39.03 received

14.

consistent with = 120 dB at 2 km

Scaling the distance to 1 km

$$= 128 \text{ dB}$$

15.

$$\text{Total Capacity} = \text{cell calls / cell} = 250 \times 6$$

$$= 1500$$

Active calls supp = arr rate $\times$ avg dur
 $= 4000 \text{ calls} \times 180 \text{ sec}$
 $= 720000$

17. Total bandwidth = channels $\times$ bandwidth / channel
 $= 150 \times 25 \text{ kHz} = 3750 \text{ kHz}$
 $= 3.75 \text{ MHz}$

18. wanted Bandwidth = $29 \times 40 \text{ kHz}$
 $= 1160 \text{ kHz}$
 $= 1.16 \text{ MHz}$

19. $s = 90 \text{ km/h}$ cell = $2 \times 15 \text{ km} = 3000 \text{ m}$
 $3000 / 25 = 30$

20. Total users = regional centers $\times$ base stations / center
 $\times$ users / base station = $8 \times 40 \times 250 = 8000$

21. App cell are = $R^2 = (0.4 \text{ km})^2$
 $= 0.16 \text{ km}^2$. Number of cells?
 $\$ 400 / 0.16 = 2500$

22. Channels = Total Spectrum / channel width
 $= 6000 \text{ kHz} / 30 \text{ kHz} = 200$

23. call minutes = $1200 \times 4 \times 2.5 = 12000 \text{ min/hr}$
 $12000 / 60 = 200$

24. users = Total bandwidth / band per user
 $= 4000 \text{ kHz} / 40 \text{ kHz} = 100$

25.
$$\text{Total capacity} = \text{users/cells} \times \text{cells} = 600 \times 1200$$

$$= 720000$$

26.
$$\text{paging operations} = \text{Total cells} / (\text{cells per operation})$$

$$= 200 / 4 = 50$$

27.
$$\text{offered} = \text{arrival rate} / \text{cell du}$$

$$= 0.17$$

28.
$$\text{users} = 157 \times 600,000 = 94,000$$

29.
$$\text{peak traffic} / \text{peak rate} = 9,000 / 12,000$$

$$= 0.75$$

$$= 75\%$$

30. Band for channels

$$48000 / 200 = 240 / 4$$

$$= 60$$